Adam Nelson

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PROFESSIONAL SUMMARY

Systems software engineer and technical leader with more than five years at AMD spanning **agentic AI tooling, MCP and RAG workflows, developer automation, and C/C++ systems**. Built an AI skills registry for more than 200 engineers associated with a 40% department productivity gain, automated 70% of test-case creation, and created an AI-tool LMS for more than 2,000 employees.

CORE COMPETENCIES

AI Tooling & Agents | MCP & RAG Workflows | Developer Enablement | Python & Bash Automation | CI/CD | C/C++ Systems | Architecture Review | Technical Leadership

WORK EXPERIENCE

AMD

July 2026 - Present

Section Manager, Software Development

Markham, ON

- Lead the Display Diagnostics team, building on prior technical ownership of its validation workflows and delivery priorities.
- Manage 11 direct reports and provide dotted-line leadership to 5 additional engineers.
- Lead an AI-based skills registry initiative in a shared repository, developing skills, MCP integrations, and RAG workflows for more than 200 engineers; increased overall department productivity by 40%.
- Created an internal learning management system for AI tools serving more than 2,000 employees across the organization.

AMD

July 2023 - June 2026

Senior Software Engineer

Markham, ON

- Architected and implemented agentic AI tools that automate 70% of test-case creation from register fields.
- Developed Python and Bash automation suites that reduced test execution time by 40%, accelerating regression cycles and feature delivery.
- Selected for the Framework Certified Code Reviewer group, responsible for architectural integrity and mentoring engineers on framework practices.
- Led verification of hardware-accelerated DMA for high-performance data-center GPUs, including the Mi450, ensuring data integrity and throughput at scale.
- Verified next-generation GPU and APU hardware from pre-silicon design through post-silicon production for DisplayPort 2.1 and HDMI 2.1 compliance using C and C++.

AMD

May 2021 - June 2023

Firmware Engineer

Markham, ON

- Developed firmware for high-efficiency video encoding and decoding, including AVC, HEVC, and VP9, for next-generation gaming consoles.
- Chaired weekly synchronization meetings for the Display Diagnostics team, aligning hardware designers and software leads to unblock critical-path work for console launches.

AMD

May 2019 - August 2020

Software Engineer Intern

Markham, ON

- Developed an Android application in Java for internal hardware validation, stress-testing video encoding, decoding, and 3D rendering pipelines.
- Created Jenkins CI/CD pipelines that reduced feature deployment time from one week to two days.

PROJECTS

Strain Characterization Apparatus - McMaster Capstone

Partnered with MesoMat to design a precision sensor-testing device with plus-or-minus 0.5% precision, earning second place among university capstone projects.

Embedded systems, data collection, cloud database synchronization

EDUCATION

Bachelor of Engineering, Software McMaster University 2021

Hamilton, ON

SKILLS

AI & Automation: Agentic AI, MCP, RAG, Python, Bash, Jenkins, n8n, LLM productivity tools

Developer & Systems: C++, C, firmware, DMA, CI/CD, Git, architecture review

Engineering: Java, Android Studio, Perforce, SQL databases, Jira, Confluence

Leadership: people leadership, team development, technical strategy, cross-functional delivery